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Cyclic Shift Phase Rotation: An Innovative, Low-Complexity Technique for PAPR Reduction

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***Abstract*—This paper presents cyclic shift phase rotation (CSPR), an innovative technique designed to mitigate the peak-to-average power ratio (PAPR) in coherent optical orthogonal frequency division multiplexing (CO-OFDM) systems. CSPR demonstrates notable improvements when compared to traditional techniques, such as discrete Fourier transform spread (DFT-spread) and selective mapping (SLM). CSPR eliminates the need for side information, additional inverse fast Fourier transforms (IFFTs), and complex search processes that are needed by other techniques. Analytical evaluations confirm that CSPR requires $O(N \log C_0)$ operations, which is significantly lower than the $O(M \cdot N \log N)$ complexity of SLM and the $O(L \cdot n \log n)$ complexity of DFT-spread. The CSPR has been evaluated within the CO-OFDM transmission. At a distance of 800 km, CSPR facilitates a 3 dB reduction in the required optical signal-to-noise ratio (OSNR) necessary for achieving hard-decision forward error correction (H-FEC) when compared to DFT-spread utilizing 112 sub-bands, and it exceeds the performance of the SLM at low OSNR with the use of 64 candidates. Moreover, it exhibits an 8 dB enhancement in the Q-factor over transmission distances extending up to 2000 km under nonlinear conditions. These results position CSPR as an effective and efficient solution for PAPR mitigation, significantly improving system performance, and reducing hardware complexity.**

***Index Terms*—PAPR, SLM, DFT-spread, nonlinear distortion, computational complexity, coherent optical OFDM, optical fiber communication**

I. Introduction

ORTHOGONAL frequency-division multiplexing (OFDM) is a widely used multi-carrier modulation format. For long-haul fiber optic transmission, coherent optical OFDM (CO-OFDM) emerges as a strong candidate due to its high spectral efficiency and ability to effectively compensate for chromatic dispersion and polarization-mode dispersion [1], [2]. However, a high peak-to-average power ratio (PAPR) is one of the major challenges that arises when utilizing OFDM in any communication system. This high PAPR pushes critical components such as IQ modulators and amplifiers into nonlinear operating regions, leading to signal distortion and power inefficiency. These nonlinear effects contribute to performance degradation by increasing error rates and reducing transmission quality. In particular, in the field of optical fiber communication, this high PAPR pushes the IQ modulator, amplifiers, and optical fiber into regimes with significant nonlinearities, which results in degrading system performance [3], [4].

Existing PAPR reduction methods, including clipping, selective mapping (SLM), and discrete Fourier transform spread (DFT-spread), each come with drawbacks such as increased computational complexity or side information requirements, limiting their practical implementation. To address the limitation of existing techniques, a novel PAPR reduction technique named cyclic shift phase rotation (CSPR) is introduced in this paper to avoid (1) transmission of side information, (2) use of extra IFFT/DFT operations on the transmitter side, (3) search complexity, (4) destroying the transmitted signals and (5) lowering spectral efficiency.

The main contributions of this paper are as follows.

1) It introduces the novel CSPR technique for PAPR reduction, specifically examined in coherent optical OFDM systems, addressing limitations of conventional methods such as side information requirements and computational inefficiencies,
2) It provides a detailed complexity analysis, demonstrating that CSPR achieves a lower computational burden $O(N \log C_0)$ compared to SLM and DFT-spread techniques, while eliminating the need for side information,
3) It evaluates the robustness of CSPR in mitigating fiber nonlinearity and avoiding nonlinear-region of IQ modulator, establishing its applicability to long-haul optical transmission systems. These contributions collectively position CSPR as a high-performance, computationally efficient solution for future optical networks.

The rest of this paper is structured as follows: Section II presents an overview of related work, detailing existing PAPR reduction techniques and their limitations. Section III outlines the conventional PAPR techniques used as a baseline reference. Section IV introduces the proposed CSPR technique, explaining its working principles and theoretical foundation. A detailed complexity analysis that compares CSPR with traditional methods is provided in Section V. Section VI describes the system model and simulation environment used to evaluate performance. Section VII discusses the simulation results, highlighting the advantages of CSPR in mitigating PAPR while maintaining low complexity. Finally, Section VIII concludes the paper and suggests directions for future research.

II. Related Work

Various PAPR reduction techniques have been investigated. The simplest technique involves amplitude clipping, where the peak envelope of the transmitted signal is limited, which leads to signal distortion by introducing in- and out-band noise. In

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[5] a new clipping and filtering method has been proposed to avoid out-of-band frequency components of clipping noise. However, the clipping method typically leads to signal distortion due to in-band noise, which significantly degrades the bit error rate (BER).

This motivates the exploration of alternative techniques to prevent signal distortion. In [6], a novel SLM scheme was proposed for OFDM non-orthogonal multiple access (NOMA) systems with reduced computational complexity while maintaining performance compared to conventional SLM. Similar work has been conducted in [7] for an optical NOMA based 5G system. In [8] an optimization of the phase factor in SLM has been investigated showing how to approximately solve it using simple linear integer programming. For CO-OFDM, a novel improved (NI-SLM) has been proposed in [9], in which no side information is introduced. Despite their effectiveness in reducing PAPR, they are computationally demanding due to the need to extra IFFT operations, search complexity, and transmission of side information.

The DFT-spread has been shown to be effective in reducing PAPR and mitigating nonlinearity of the fiber in [10]–[12]. This is due to spreading the energy of data symbols across multiple subcarriers before the IFFT operation, it lowers the peak power of the transmitted signal. Although this technique avoids the complexity associated with the search process and side information, it relies heavily on a considerable number of DFT operations that are computationally demanding. The work in [13] investigates the partial transmit sequence (PTS) technique based on particle swarm optimization (PSO) for the reduction of PAPR in OFDM based on wavelet packets, focusing on balancing complexity and performance. However, it requires transmitting of side information to the receiver, along with involving multiple levels of filtering and decomposition where additional IFFTs are likely required. In [14] a tone reservation method has been explored with a minimized ℓ_∞ norm for the PAPR reduction in OFDM signals, to achieve an improved peak cancellation effectiveness, where a subset of subcarriers (tones) is reserved and is unused for data transmission. These reserved tones are used to cancel the peak power components in the original signal. Although neither side information nor extra FFTs are required, it leads to a reduction of the spectral efficiency.

Another notable approach is the phase rotation method within a mixed-numerology architecture for PAPR reduction in 5G and beyond [15]. Unlike conventional single-numerology OFDM, this technique exploits the flexible numerology structure of 5G to introduce novel phase rotation strategies. This method eliminates the need for additional IFFTs and side information transmission, reducing computational overhead compared to the SLM technique. However, optimization of phase factors still requires a search process and the symbol-based approach demands a larger number of phase search iterations, increasing complexity.

Companding techniques are widely used to mitigate PAPR. These nonlinear techniques compress high peaks and amplify low amplitudes in the signal to reduce PAPR. One of the companding techniques is the root-based nonlinear companding (RMC) [16] which employs root functions to manage the amplitude distribution, thereby achieving a reduction in PAPR with minimal impact on BER. In contrast, the piecewise linear companding (PLC) approach [17] provides an effective PAPR reduction with a lower complexity. However, this method introduces an error floor for high-order QAM at low PAPR target values. Due to the application of a nonlinear function, companding techniques inherently introduce nonlinear distortion, which can adversely affect BER performance. Furthermore, these techniques contribute to spectral regrowth, resulting in increased out-of-band interference and necessitating additional filtering [18].

Deep learning has also been used for PAPR reduction, as seen in recent work on convolutional autoencoder (CAE)-based MIMO-OFDM waveform design [19]. This approach integrates PAPR reduction with MIMO detection using a joint learning framework. While eliminating the need for side information and allowing end-to-end optimization, the method introduces significant computational overhead due to neural network training and inference. The iterative nature of the MIMO detection block further increases the processing complexity, which may limit its applicability to real-time systems with stringent latency requirements.

Given the challenges of existing techniques, this work presents CSPR as a novel and computationally efficient approach without requiring additional IFFT operations, side information transmission, complex search processes, lowering spectral efficiency, or introducing BER degradation. With a computational complexity of $O(N \log C_0)$, the CSPR provides a superior trade-off between performance and complexity, making it a practical choice for CO-OFDM systems.

III. Traditional PAPR Reduction Techniques

Traditional PAPR techniques serve as baselines to evaluate the performance of CSPR reduction. Clipping, DFT-spread, and SLM are considered for comparison.

A. Clipping method

The technique of clipping represents the most straightforward method of reducing PAPR. The fundamental concept is based on the limitation of the amplitude by constraining the peak envelope of the transmitted signal. Although this approach results in signal distortion perceived as noise, it does not present computational complexity. This noise appears both in-band and out-of-band. Out-of-band distortions can be reduced by filtering out, but it needs repeated clipping-and-filtering, while in-band distortion cannot be reduced. The clipping threshold is calculated as $T_{clip} \leftarrow P_{clip} \cdot \max_i |x|$, where P_{clip} is the percentage of the maximum magnitude used for clipping, and x are the time-domain signal data. The clipped data $\hat{x}_i$ is determined as [20]:

$$\hat{x}_i = \begin{cases} T_{clip} \cdot e^{j\angle(x_i)} & \text{if } |x_i| > T_{clip} \\ x_i & \text{otherwise} \end{cases} \tag{1}$$

where i is the index of the time-domain signal data, $\angle(x_i)$ is the phase of the i-th data in radians, to maintain the original phase.

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B. DFT-spread

In DFT-spread OFDM, the transmitter initially divides the data into L sub-bands as shown in Fig. 1 [21]. Each sub-band undergoes a DFT operation, followed by an IFFT operation applied to the combined sub-bands to generate a time-domain signal. This process effectively reduces PAPR by distributing the energy of each data symbol across a larger frequency band, thereby lowering the peak power of the transmitted signal.

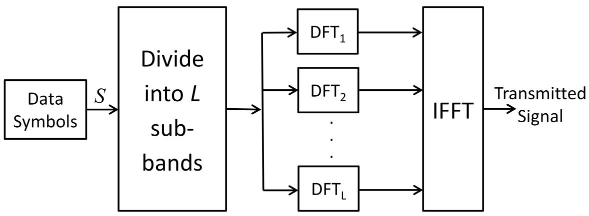

Fig. 1: A block diagram of DFT-spread

Although DFT-spread is effective in PAPR reduction, it exhibits significant sensitivity to laser phase noise induced by high linewidth [22]. This comes from the influence between neighboring sub-bands just as among the subcarriers of conventional OFDM with narrow spacing.

C. Selective Mapping (SLM)

SLM is another technique used to reduce PAPR, which does not cause signal distortion. As shown in Fig. 2, SLM technique generates M candidate data symbols C^m, $1 \leq m \leq M$ by multiplying input data symbols S of N-length with M phase sequence candidates $B^m = [B_0^m, B_1^m, ..., B_{N-1}^m]$ where B^m is an N-length vector containing elements from the set $\{1, -1, j, -j\}$ as [23]

$$C^m = B^m \cdot S, \tag{2}$$

the symbol with the lowest PAPR is selected for transmission as

$$\widehat{C} = \underset{1 \leq m \leq M}{argmin} \left(\frac{\max\limits_{0 \leq n \leq N-1} [|C^m|^2]}{ave\left[|C^m|^2\right]} \right), \tag{3}$$

where $max[\cdot]$ and $ave[\cdot]$ are the maximum and average of data symbol power of N subcarriers. The phase sequence $\widehat{B}$ associated with the lowest PAPR must be transmitted as side information to accurately reconstruct the data in the receiver. Additionally, implementing this technique requires multiple IFFT blocks equal to M phase sequence candidates that increase transmitter complexity.

The main steps of the SLM technique are the following: First, the algorithm generates candidate symbols for each phase sequence, by multiplying the original OFDM symbol with the phase sequences; then it performs the n-point IFFT operation to convert the symbol to the time domain and calculate the PAPR. Accordingly, the best symbol and the best phase sequence are updated if the current PAPR is lower than the best PAPR found so far. After examining all the candidate symbols, the algorithm returns the OFDM symbol with the lowest PAPR and its phase sequence that needs to be transmitted to the receiver.

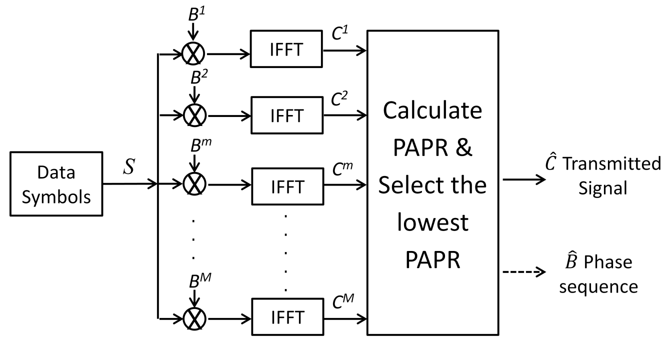

Fig. 2: A block diagram of SLM technique

IV. The Proposed Cyclic Shift Phase Rotation (CSPR)

The CSPR algorithm iteratively applies phase rotations. As shown in Fig. 3, first the original OFDM symbol data undergo element-wise multiplication with the corresponding predefined phase sequence. Then the phase vector is cyclically shifted by C_k positions after each iteration k, starting with an initial shift amount of C_0. This process is crucial to ensure that the phase vector changes for each iteration and to avoid applying the same phase rotation repeatedly. The cumulative effect of these rotations spreads the signal energy across subcarriers, reducing peaks in the time domain.

The CSPR technique is given by:

$$\hat{S}^{(k)} = \hat{S}^{(k-1)} \cdot P^{(k)}, \tag{4}$$

where $\hat{S}^{(k)} = [\hat{s}_0^{(k)}, \hat{s}_1^{(k)}, \ldots, \hat{s}_{N-1}^{(k)}]$ represents the phase rotated OFDM symbol in frequency domain at the k-th iteration stage, N is the number of subcarriers, and $P^{(k)} = \left(\exp\left(j\Phi^{(k)}\right)\right)$ is the phase rotation vector applied at the k-th iteration stage, with $\Phi^{(k)} = [\phi_0^{(k)}, \phi_1^{(k)}, \ldots, \phi_{N-1}^{(k)}]$ representing the phase sequence applied at the k stage. Consequently, the phase rotation $P^{(k)}$ only modifies the phase of the symbols. In accordance with $S = |S| \cdot e^{j\theta}$, the symbols subjected to rotation at the k-th iteration are described as

$$\hat{S}^{(k)} = |\hat{S}^{(k-1)}| \cdot e^{j(\theta + \Phi^{(k)})}. \tag{5}$$

$K = \log_2(C_0)$ is the total number of iteration stages, and C_0 is the initial cyclic shift size. The underlying principle is that at each stage k, the algorithm applies a different phase shift to the symbol data. The output $\hat{S}^{(K)}$ is the phase-rotated frequency-domain signal that minimizes the PAPR when converted to the time domain using the IFFT.

$$\hat{s}^{(K)}[t] = \text{IFFT}(\hat{S}^{(K)}). \tag{6}$$

In summary, CSPR iteratively adjusts phases, ensuring that subcarriers do not align to form peaks. Instead of a specific phase vector like in SLM, CSPR applies a phase vector that undergoes cyclic shifts in each iteration. This way ensures that each iteration spreads different phase shifts among subcarriers in each iteration, thereby maximizing phase diversity and reducing PAPR. This cyclic shift facilitates an optimal phase distribution that prevents high peaks from forming. This

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process is computationally efficient, requiring no additional IFFTs, no search complexity, or side information transmission. Moreover, as the predefined phase sequence is known, it is fully reversible at the receiver, where the original signal is recovered by applying the inverse of the same phase sequence in reverse order.

The following outlines a step by step procedure of the proposed CSPR.

1) Initialization: start with the original data of the OFDM symbol S and define the initial cyclic shift size C_0 and the phase rotation vector $P^{(1)}$ for the first iteration.
2) Phase multiplication: apply element-wise multiplication of the original symbol data with the predefined phase sequence, resulting in a phase-rotated signal $\hat{S}^{(1)}$.
3) Cyclic shift: after each iteration, the phase vector is cyclically shifted by C_k positions (C_0 is used in the first iteration), where k is the current iteration number. This ensures that different phase rotations are applied in subsequent iterations.
4) Iteration: repeat the phase multiplication and cyclic shift process for a total of $K = log_2(C_0)$ iterations, halving the size of the cyclic shift ($C_{k+1} = C_k/2$) at the end of each iteration until it reaches 1.
5) Output generation: the final output $\hat{S}^{(K)}$ is the phase-rotated frequency domain signal, which minimizes the PAPR when converted back to the time domain using the IFFT.
6) Reversibility: The predefined phase sequence allows the original signal to be recovered at the receiver by applying the inverse of the phase sequence in reverse order.

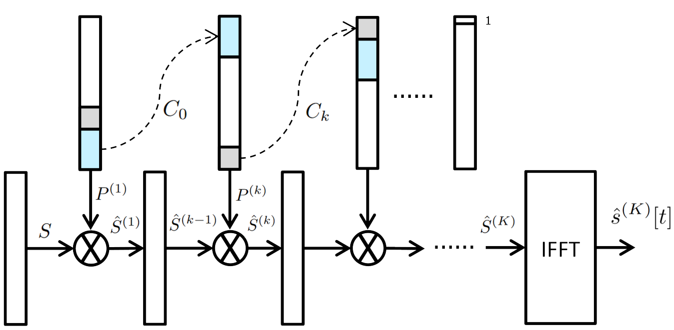

Fig. 3: A block diagram of the proposed CSPR technique

A. CSPR Algorithm Pseudocode

Algorithm 1 presents the pseudocode of the proposed CSPR. The algorithm takes as input a $signal$, an $initial\ cyclic\ shift$ (C_0) size and an optional set of phases. The $processed\_signal$ and $phases$ are the output of the algorithm. If no phases are provided, the algorithm generates random phases of signal length. The algorithm iteratively applies phase rotations $phases$ sequence to the $processed\_signal$ using *ApplyPhase* function. This function performs element-wise multiplication between the vector $processed\_signal$ and $e^{(j \cdot phases)}$. At the end of each iteration, the $phases$ sequence is cyclically shifted by $cyclic\ shift$, given the size of the cyclical shift. In the end, the $cyclic\ shift$ size is halved and the algorithm continues to the next iteration until the cyclic shift size reaches 1. Finally, $processed\_signal$ is returned as the output of the algorithm, along with the $phases$ vector used for rotation.

Algorithm 1 CSPR Algorithm

Require: Signal $signal$, Initial Cyclic Shift $initial\_cyclic\_shift$ (C_0), optional set of phases $phases\_in$
Ensure: Processed signal $processed\_signal$
$cyclic\_shift \leftarrow initial\_cyclic\_shift$
$processed\_signal \leftarrow signal$
if $phases\_in$ is None **then**
  $phases \leftarrow$ generate_random_phases($signal\_size$) ▷ Generate random phases
else
  $phases \leftarrow phases\_in$
end if
while $cyclic\_shift > 1$ **do**
  $processed\_signal \leftarrow$ ApplyPhase($processed\_signal, phases$) ▷ Apply phase rotation to signal
  $phases \leftarrow$ CyclicShift($phases, cyclic\_shift$)
  $cyclic\_shift \leftarrow cyclic\_shift/2$
end while
return $processed\_signal, phases$

An example clarifying the CSPR algorithm is provided below, under the assumption that four subcarriers are utilized for the sake of simplicity.

1) Initialize a random phase sequence selection from an array of potential phase values, resulting in $\Phi = [\pi/2, \pi, 3\pi/2, 0]$,
2) the application of phase rotation to the original OFDM symbol $S = [s_0, s_1, s_2, s_3]$ is conducted in the frequency domain through element-wise multiplication with the generated phase value, yielding the result as outlined in $\hat{S} = \left[s_0 e^{j\pi/2}, s_1 e^{j\pi}, s_2 e^{j3\pi/2}, s_3 e^{j0}\right]$,
3) cyclically shift the phase sequence by C_0, assuming two positions, the new phase sequence becomes $\Phi^1 = [3\pi/2, 0, \pi/2, \pi]$,
4) consequentially the new rotated symbols become $\hat{S}^1 = [\hat{s}_0 e^{j3\pi/2}, \hat{s}_1 e^{j0}, \hat{s}_2 e^{j\pi/2}, \hat{s}_3 e^{j\pi}]$,
5) the cyclic shift size is halved, followed by a reapplication of phase rotation, until the cyclic shift size is reduced to a value of one.

B. CSPR Mathematical Model

1) Initial Signal Representation: The original symbol in the frequency domain is represented as $S = [s_0, s_1, \ldots, s_{N-1}]$, where N is the number of subcarriers. A predefined phase sequence is denoted by $\Phi = [\phi_0, \phi_1, \ldots, \phi_{N-1}]$. ϕ_i is a continuous value drawn from the range $[0, 2\pi)$ or a value selected from a finite set of $\{0, \pi, \pi/2, 3\pi/2\}$. The initial cyclic shift is defined as $C_0 \in 2^K$ where K is the number of iterations. The goal is to cyclically shift Φ by C_k, so that the phase vector changes for each iteration.

TABLE I: Comparison of CSPR, SLM, and DFT-spread techniques

Steps	**CSPR**	**SLM**	**DFT-spread**
Main Operations	iteratively apply phase rotation	IFFT for each candidate symbol, phase sequence multiplication	DFT operation for each sub-bands
Number of Iterations	$\log_2(C_0)$	M (number of candidate symbols)	L (number of sub-bands)
Complexity per Iteration	$O(N)$	$O(N \log N)$ (IFFT)	$O(n \log n)$ (DFT)
Total Complexity	$O(N \log C_0)$	$O(M \cdot N \log N)$	$O(L \cdot n \log n)$
Side Information	None	Needs phase sequence transmission	None
Additional IFFTs	None	M IFFTs	L DFTs
Search Complexity	None	Yes $O(M)$	None
Advantages	Effective PAPR reduction, Lower complexity	Effective PAPR reduction, no signal distortion	Efficient computational complexity compared to SLM
Disadvantages	Requires iterative processing	Higher complexity due to multiple IFFTs, search complexity and phase sequence transmission	Requires L $\cdot$ DFT operations

2) Iteration Multiplication Process: In each iteration k, the signal $\hat{S}$ is updated by applying the corresponding phase vector as

$$\hat{S}^{(k)} = \hat{S}^{(k-1)} \cdot e^{j\Phi^{(k)}}, \tag{7}$$

3) Cyclic Shift Phase Rotation: At the end of each iteration k, the phase vector Φ^k is cyclically shifted by C_k (the initial value is C_0). This ensures that the phase vector changes for each iteration.

$$\phi_n^{(k+1)} = \phi_{(n+C_k) \mod N}^{(k)}, \tag{8}$$

where $\phi_n^{(k+1)}$ is the phase shift applied to subcarrier n at iteration k and the modulus operation (mod N) ensures that the phase shifts wrap around cyclically when reaching the end of the N subcarriers. Finally, the cyclic shift C_k is halved for the next iteration $C_{k+1} = C_k/2$, the iteration process ends at $C_k = 1$, when the cyclic shift reduces to the smallest possible size.

4) Receiver Process: At the receiver, the inverse phase rotation is applied in reverse order to recover the original signal as

$$\hat{R}^{(k)} = \hat{R}^{(k-1)} \cdot e^{-j\Phi^{(k)}}, \tag{9}$$

where $R^{(k)}$ represents the received signal at the k-th iteration. In order to integrate the proposed CSPR into the mathematical representation of the OFDM signal, assuming the expression of the discrete-time baseband OFDM signal is given as [24], [25]

$$s[t] = \sum_{n=0}^{N-1} |S_n| e^{j(\frac{2\pi nt}{N} + \theta_n)}, \text{ for } t = 0, 1, \ldots, N-1\ , \tag{10}$$

where $|S_n|$ is the magnitude of the transmitted symbol corresponding to the n-th subcarrier in the frequency domain, θ_n represents the phase associated with the n-th subcarrier, t is the discrete time index, and $2\pi n/N$ represents the frequency of n-th subcarrier. In scenarios of high PAPR, subcarriers tend to have similar phases (aligned phase coherence), causing peaks in $s[t]$. While the application of phase rotations could theoretically introduce new phase alignments that increase PAPR, the iterative and cyclic shift design of CSPR ensures that such scenarios are statistically negligible.

The CSPR technique modifies the OFDM signal by applying a phase rotation sequence iteratively, as outlined in Steps (2) and (3). Consequently, the signal power after K iterations is:

$$|\hat{s}^{(K)}[t]|^2 = \left| \sum_{n=0}^{N-1} |S_n| e^{j(\frac{2\pi nt}{N} + \theta_n + \sum_{k=0}^{K-1} \phi_{n+C_k}^{(k)})} \right|^2. \tag{11}$$

This way establishes uncorrelated phases by adding $\sum_{k=0}^{K-1} \phi_{n+C_k}^{(k)}$, which effectively reduces peak occurrences and consequently results in a reduced PAPR.

V. Computational Complexity

A. DFT-spread Complexity Analysis

The computational complexity of DFT-spread in the transmitter can be analyzed as follows.

- Division into sub-bands: The OFDM symbol is divided into L disjoint sub-bands. This involves simple indexing and does not significantly contribute to the overall complexity.
- DFT operation: Each sub-band undergoes DFT operation which has a complexity of $O(n \log n)$, where n is the input subcarriers to the DFT operation, that is $n = N/L$.
- Number of sub-bands: Since there are L sub-bands, the total complexity is $O(L \cdot n \log n)$.

The complexity of DFT-spread arises from the DFT operations, therefore the total big O notation is $O(L \cdot n \log n)$, where L is the number of sub-bands.

B. SLM Technique Complexity Analysis

To determine the big O notation for the SLM technique, we analyzed its computational complexity according to the following steps:

- Generation of candidate symbols: The SLM technique generates M candidate OFDM symbols by multiplying the original OFDM symbol with M different phase sequences. Each multiplication involves N operations, where N is the number of subcarriers. Therefore, the complexity of generating candidate symbols is $O(M \cdot N)$.

- IFFT operation: For each candidate OFDM symbol, an IFFT operation is performed. The complexity of a N-point IFFT is $O(N \log N)$. Since there are M candidate symbols, the total complexity for IFFT operations is $O(M \cdot N \log N)$.
- Selection of the best candidate: The SLM algorithm selects the candidate with the lowest PAPR. This involves calculating the PAPR for each candidate, which is $O(N)$ for each candidate. Therefore, the complexity of selecting the best candidate is $O(M \cdot N)$.

Combining these steps, the overall time complexity of the SLM technique is $O(M \cdot N + M \cdot N \log N + M \cdot N)$, By simplifying this, we get $O(M \cdot N \log N)$

C. Proposed CSPR Technique Complexity Analysis

We analysis the computational complexity of the proposed CSPR based on Algorithm 1 by determining the big O as follows:

- Iterative multiplication: The principle computational endeavor arises from applying phase rotations iteratively. For each iteration, the algorithm performs element-wise multiplications, which is an operation $O(N)$, where N is the number of input elements (number of subcarriers).
- Cyclic shifting and halving: At the end of each iteration, the cyclic shift is applied to the phases sequences. This can be done by slicing and concatenating the phase vector, which involves creating two slices of sizes N in total and then concatenating these slices, which leads to a complexity of $O(N)$. The size of the cyclic shift is then halved, which gives a constant-time operation $O(1)$.
- Number of iterations: The number of iterations is determined by the initial size of the cyclic shift C_0 and the halving process until it reaches 1. Thus, the total number of iterations is $\log_2(C_0)$.
- Total complexity per iteration: In each iteration, the algorithm processes all elements of the signal. Therefore, the total complexity is $O(N + 1 + N)$, which results in $O(N)$.
- Overall complexity: Considering $\log_2(C_0)$ iterations, the total complexity is given by $O(N \cdot \log_2(C_0))$.

To sum up, the complexity of the CSPR algorithm for each iteration is $O(N)$. This complexity arises because the algorithm performs $O(N)$ operations in each of the $O(\log C_0)$ iterations. Therefore, the CSPR algorithm has a complexity of $O(N \cdot \log C_0)$, which is more effective than the other techniques that require additional complex operations such as IFFT operations, DFT operations, search process, or side information transmission.

D. Comparison Analysis

Table I presents the complexity comparison of the three algorithms. The SLM has a complexity of $O(M \cdot N \log N)$, where M is the number of candidate symbols. Thus, in addition to the search process and the transmission of phase sequences, it requires M instances of IFFT, leading to higher complexity. The DFT-spread requires $O(L \cdot n \log n)$, where L is the number of sub-bands, which is equivalent to the number of DFTs required. The main advantage of DFT-spread is that it is efficient in terms of computational resources, as there is no side information or search process, and $n < N$. Finally, the proposed CSPR technique has a complexity of $O(N \log C_0)$. It avoids the need for extra IFFTs/DFTs, search process, and side information, which is more efficient in terms of computational complexity and implementation.

A comparative example of computational complexity is provided in Table II. Taking into account the use of parameters $N = 1024$ and $C_0 = 4$, the iteration number (K) is determined to be $log_2(C_0) = 2$ for the proposed CSPR approach. For the SLM, 64 candidates are used, and in the case of DFT-spread, 112 sub-bands are utilized. The complexity per iteration and the total complexity for each technique are calculated as in Table II.

TABLE II: Example of Complexity Comparison

Technique	Complexity per Iteration	Total Complexity
CSPR	$O(N) = 1024$	$O(N \log_2 C_0) = 1024 \times 2 = 2048$
SLM	$O(N \log N) = 1024 \times 10 = 10240$	$O(M \cdot N \log N) = 64 \times 10240 = 655360$
DFT-spread	$O(n \log n) = 9 \times 3 = 27$	$O(L \cdot n \log n) = 112 \times 27 = 3024$

VI. System Model

The OFDM coder and decoder, which incorporate PAPR reduction techniques, are implemented in Python, as shown in Fig. 4. After serial to parallel (S/P) operation, the pseudorandom binary sequence (PRBS)-31 bit stream is mapped to a 16-QAM constellation, followed by CSPR (or SLM, or DFT-spread). Such higher-order modulation formats generally lead to higher PAPR. Oversampling is necessary to spectrally isolate the OFDM baseband signal from high-frequency aliasing products, therefore, 128 zero subcarriers are inserted in the middle of IFFT input. The RF-pilot method with -20 dB pilot-to-signal ratio (PSR) and a 40-subcarrier guard band (GB) is included to mitigate laser phase noise [26], [27] (see Appendix A). The resulting frequency domain signal illustrated in Fig. 5 is converted to the time domain using a 1024-point IFFT.

Over 2000 km, the required cyclic prefix (CP) is approximately 0.8 ns as given by [28]:

$$CP_{required} \geq \frac{c}{f_c^2} \cdot CD_{todal} \cdot N \cdot \Delta f \ , \qquad (12)$$

where c represents the speed of light, f_c represents the frequency of optical carrier (193.1 THz), CD_{total} is the total chromatic dispersion over transmission link and Δf represent the subcarrier spacing. Therefore, 64 subcarriers for cyclic prefix, which results in 3.2 ns, are more than sufficient to mitigate inter-symbol interference (ISI) caused by chromatic dispersion. The generated baseband OFDM signal is then modulated onto an optical carrier using an IQ Mach-Zehnder Modulator (modeled based on Appendix B) with a laser linewidth of 10

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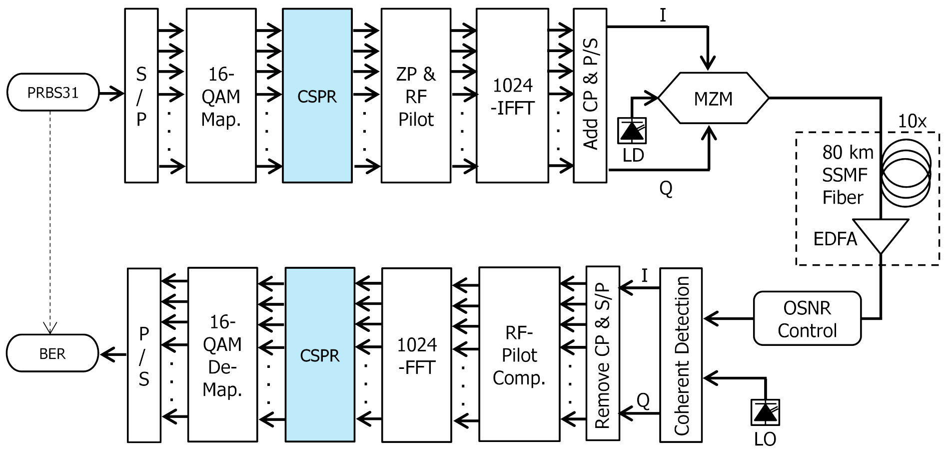

Fig. 4: CO-OFDM system transmission setup

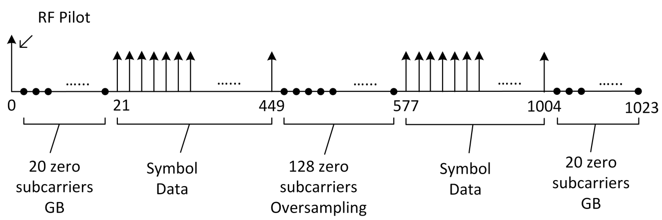

Fig. 5: 1024-point IFFT subcarrier assignment

kHz (modeled as in Appendix C). Such a narrow linewidth has been set to mainly study the impact of nonlinearites of the IQ-modulator and fiber. The transmission link (modeled in Appendix D) consists of several spans, each including 80 km standard single-mode fiber (SSMF) and erbium-doped fiber amplifier (EDFA) amplifiers of 4.5 dB noise figure level to compensate for 0.2 dB/km fiber loss. Consequently, the realization of a transmission of 800 km requires the use of 10 spans. The fiber chromatic dispersion and fiber nonlinearity are set at 16 ps/nm/km and 1.3 $\text{W}^{-1}.\text{km}^{-1}$, respectively. Various launch powers are examined, starting from -14 dBm to -4 dBm. The simulation is carried out at a sampling rate of 2.5 GSa/s.

The coherent receiver is modeled as in Appendix E, where the optical signal is detected and converted to an electrical signal with a local oscillator of 10 kHz. The electrical signal is then passed to the OFDM decoder, where the RF-Pilot is first extracted and used to compensate for laser phase noise [27], [29]. To mitigate the effects of chromatic dispersion, which introduces a frequency-dependent phase shift, data-aided channel estimation is performed [30]. Each OFDM frame consists of 52 OFDM symbols (each OFDM symbol corresponds to 1024 points), with 4 training symbols used to accurately estimate the channel frequency response. Then a 1024-point FFT converts the time-domain signal back to the frequency domain. Finally, after inverse CSPR (or inverse SLM, or IDFT-spread), 16-QAM demapping with parallel to serial (P/S) is applied to recover the bit stream that is then compared to the original PRBS-31 bit stream to calculate the BER. The optical part of the simulation is implemented using OptiCommpy, which is an open source Python library [31].

Figure 6 illustrates the placement of the RF-Pilot within the OFDM signal spectrum at a launch power of -12 dBm. They show the power spectral density (PSD) of optical signals at different stages of transmission, with and without the proposed CSPR. Figure 6 (a) shows the optical spectrum after modulation of IQ with RF pilot without any reduction techniques. In Figure 6 (b), which is measured at the end of the 800 km fiber link, significant distortion around the GP region, the oversampling region and the OFDM signal can be observed. This distortion is primarily due to nonlinear effects in the optical fiber, which are exacerbated by high PAPR signals. Figure 6 (c) represents the scenario with the proposed PAPR reduction technique (CSPR) at the output of the IQ modulator. Compared to Fig. 6 (a), it is clearly seen that the signal is more smothered (as is obvious in the GB around RF-pilot and the oversampling region) and the impact of the nonlinear IQ modulator has been reduced due to PAPR reduction. After 800 km fiber distance and in Fig. 6 (d), the distortions around the GB, oversampling, and the OFDM signal are significantly reduced compared to Fig. 6 (b). This indicates that the CSPR technique has effectively mitigated the impact of nonlinear distortions.

VII. Performance Analyses and Results

The complementary cumulative distribution function (CCDF) of the transmitted OFDM signals is used to evaluate PAPR reduction techniques. The CCDF provides the probability that the PAPR exceeds a given threshold. The PAPR is calculated as [32]:

$$PAPR(\epsilon) := \min(v) \text{ s. t. } P\left(\frac{|s(t)|^2}{\mathbb{E}[|s(t)|^2]} > v\right) \leq \epsilon \ , \quad (13)$$

where $s(t)$ is the transmitted signals in time domain. $PAPR(\epsilon)$ is defined as the minimum value of v for which the probability of $\frac{|s(t)|^2}{\mathbb{E}[|s(t)|^2]}$ being greater than v is less than ϵ, where ϵ is a positive number less than 1. It calculates the smallest power threshold v so that the probability of instantaneous power of the signal that exceeds the threshold is

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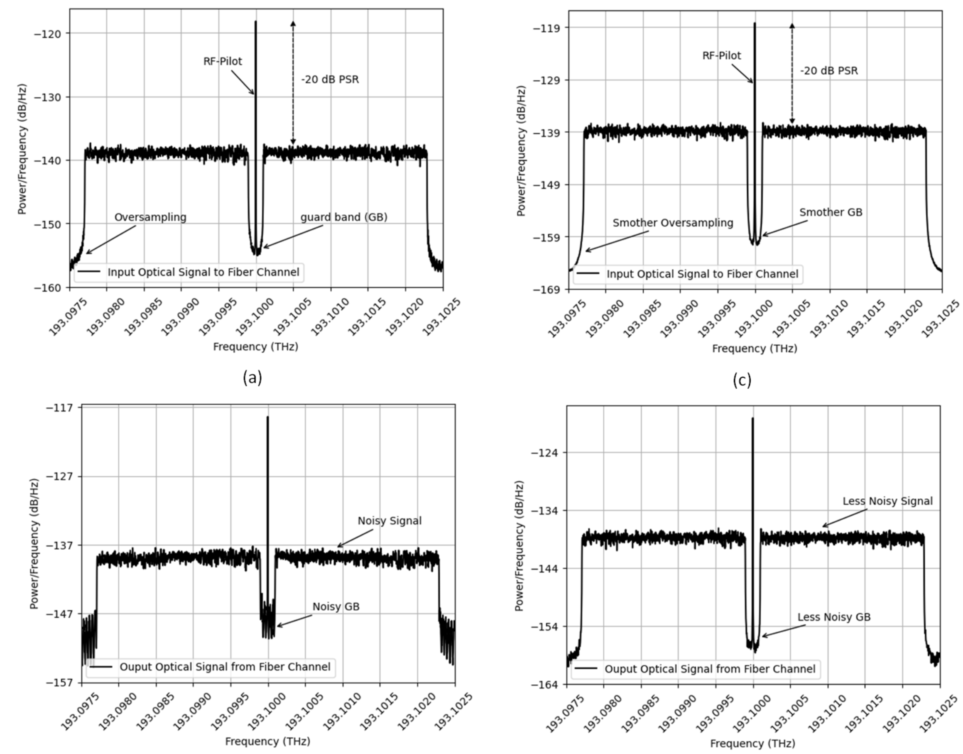

Fig. 6: Optical OFDM power spectral density at launch power of -12 dBm over 800 km (a) optical signal after IQ modulation without PAPR reduction (b) received optical signal to the coherent receiver without PAPR reduction (c) optical signal after IQ modulation with proposed CSPR (d) received optical signal to the coherent receiver with proposed CSPR

also less than or equal to ϵ. This would show a more practical measure of the peak power of the signal compared to the traditional PAPR definition that only considers the absolute maximum peak. For example, when $\epsilon = 10^{-2}$, it means that the probability that the instantaneous power of the signal is more than 100 times its average power ($\frac{|s(t)|^2}{\mathbb{E}[|s(t)|^2]} > 100$) is less than or equal to 1%.

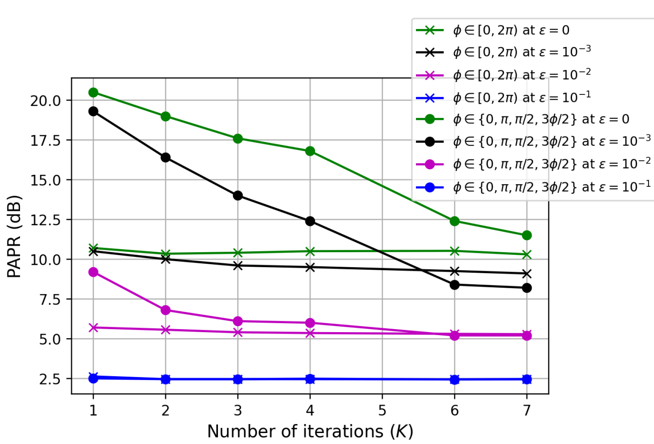

Fig. 7: Calculated PAPR at CCDF of 10^{-3} of the transmitted OFDM signals (generated usng 16-QAM with 1024-point IFFT) for the proposed CSPR as a function of iteration numbers (K)

Fig. 7 compares the PAPR reduction achieved by CSPR using different total numbers of iterations (K) for different ϵ values at CCDF of 10^{-3}. This evaluation is conducted for a 16-QAM modulation format with a 1024-point IFFT. Initially, we investigated a phase rotation using a finite set of phase shifts $\{0, \pi, \frac{\pi}{2}, \frac{3\pi}{2}\}$. The results show that by increasing the total number of iterations to 6, it leads to a significant PAPR reduction of the proposed CSPR. However, when using the continuous phase shift range of $[0, 2\pi)$, further iterations yielded only marginal improvements in PAPR reduction. Consequently, it can be observed that the PAPR reduction values are nearly identical with higher iteration numbers (K). However, as discussed in Section V, it is important to note that a higher total number of (K) also leads to increased computational complexity. A value of $K = 2$ is selected for further investigation to provide a suitable compromise between PAPR reduction and computational complexity.

TABLE III: PAPR values and reductions for different techniques at $\epsilon = 10^{-3}$

Method	PAPR value (dB)	PAPR Reduction (dB)
No reduction	∼17.3	0
Clipping (90%)	∼17.3	0
SLM (64 Candidates)	∼11.9	∼5.4
DFT-s (112 Sub-bands)	∼14.8	∼2.5
CSPR (K=2)	∼10.3	∼7

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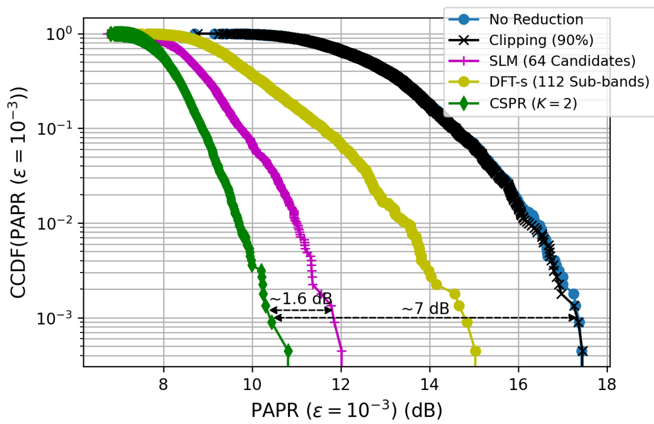

Fig. 8: CCDF of the transmitted OFDM signals with different PAPR reduction at $\epsilon = 10^{-3}$

Fig. 8 illustrates the calculated PAPR at $\epsilon = 10^{-3}$ for a 16-QAM modulation format with a 1024-point IFFT. The results demonstrate the effectiveness of the proposed CSPR technique at $K = 2$ in reducing PAPR. CSPR achieves an approximate reduction of 7 dB and outperforms SLM (64 candidates) and DFT-spread (112 sub-bands) by 1.6 dB and 4.5 dB, respectively. Table III provides a detailed numerical comparison of PAPR values and reductions for different techniques at $\epsilon = 10^{-3}$, further supporting the trends observed in Fig. 8. The results show that the baseline system without any reduction has a PAPR of approximately 17.3 dB, while conventional clipping (90%) provides negligible improvement, maintaining nearly the same PAPR level. In contrast, SLM (64 candidates) and DFT-spread (112 subbands) achieve PAPR reductions of 5.4 and 2.5 dB, respectively. However, CSPR surpasses these techniques by achieving a reduction of 7 dB, along with eliminating the need for side information, complex search processes, or additional IFFT computations. These findings confirm that CSPR provides a computationally efficient and highly effective solution for reducing PAPR while maintaining a lower computational burden.

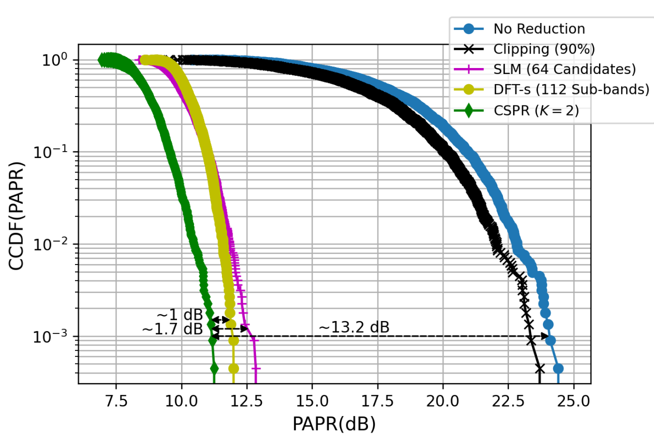

Fig. 9: CCDF of the transmitted OFDM signals with different PAPR reduction at $\epsilon = 0$

TABLE IV: PAPR values and reductions for different techniques at $\epsilon = 0$

Method	**PAPR value (dB)**	**PAPR Reduction (dB)**
No reduction	∼24.2	0.0
Clipping (90%)	∼23.5	∼0.7
SLM (64 Candidates)	∼12.7	∼11.5
DFT-s (112 Sub-bands)	∼12.0	∼12.2
CSPR (K=2)	∼11.0	∼13.2

Fig. 9 shows the CCDF of transmitted signals at $\epsilon = 0$, which is equivalent to conventional PAPR calculation where

$$PAPR = \frac{\max |s(t)|^2}{\mathbb{E}[|s(t)|^2]}. \tag{14}$$

At $\epsilon = 0$, the results show that CSPR with 2 iterations ($K = 2$) achieves a PAPR reduction of approximately 13.2 dB for the same system configuration. Additionally, the reduction of around 1 dB and 1.7 dB, compared to the DFT-spread and SLM technique, respectively. Table IV provides a numerical breakdown of the PAPR values and reductions for different PAPR reduction techniques at $\epsilon = 0$, reinforcing the insights gained from Fig. 9. It highlights that conventional clipping (90%) achieves only a marginal reduction of 0.7 dB due to distortion effects, while SLM with 64 candidates and DFT-spread with 112 sub-bands offer more substantial reductions of 11.5 and 12.2 dB, respectively. However, CSPR surpasses both techniques with a 13.2 dB PAPR reduction, demonstrating its superior performance while maintaining lower complexity.

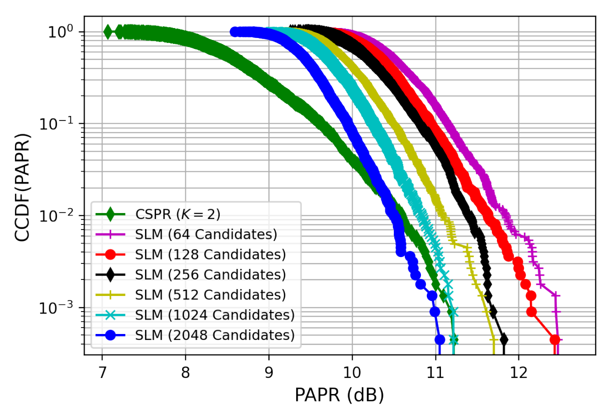

Fig. 10: Comparison of CCDF for the SLM with varying number of candidates and the proposed CSPR at $\epsilon = 0$

TABLE V: PAPR values and reductions for SLM technique with difference candidates vs. the proposed CSPR at $\epsilon = 0$

Method	PAPR value (dB)	PAPR Reduction (dB)
No Reduction	∼24.2	0.0
SLM (64 Candidates)	∼12.5	∼11.7
SLM (128 Candidates)	∼12.2	∼12
SLM (256 Candidates)	∼11.7	∼12.5
SLM (512 Candidates)	∼11.5	∼12.7
SLM (1024 Candidates)	∼11.3	∼12.9
SLM (2048 Candidates)	∼11	∼13.2
CSPR	∼11.2	∼13

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Fig. 10 compares the CCDF of the proposed CSPR technique with the SLM technique for different candidate sizes, namely 64 to 2048, at $\epsilon = 0$. It provides insights into their PAPR reduction capabilities. The results indicate that CSPR achieves a PAPR reduction of approximately 13 dB, comparable to approximately 13.2 dB for SLM with 2048 candidates, but with significantly lower computational complexity. Table V further quantifies this comparison, listing the PAPR values and the corresponding reductions for the proposed CSPR technique and the SLM techniques with different candidate numbers. It is evident that increasing the number of candidates for SLM improves the reduction in PAPR, but even with 2048 candidates, SLM only marginally exceeds CSPR by 0.2 dB, at the cost of higher complexity due to multiple IFFT computations, search complexity, and side information requirements. This highlights the efficiency of CSPR, achieving competitive PAPR reduction without extra processing, making it a viable alternative for practical implementation.

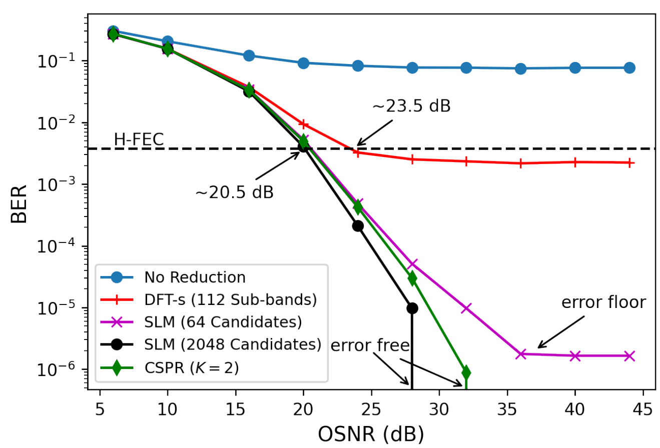

Fig. 11: BER performance vs OSNR of the received signals over 800 km at launch power of −12 dBm

Fig. 11 presents a comparison of BER performance of CSPR and baseline reduction techniques over the 800 km transmission system at a launch power of −12 dBm, as a function of the optical signal-to-noise ratio (OSNR). The required OSNR for hard-decision forward error correction (H-FEC) achieved by the CSPR is around 3 dB lower than DFT-spread with 112 sub-bands and almost identical to the required OSNR achieved by SLM. Nevertheless, the SLM technique using 64 candidates exhibits an error floor at higher OSNR, whereas the SLM technique using 2048 candidates, which requires a relatively high computational complexity and CSPR maintain error-free performance beyond 28 and 32 dB OSNR, respectively.

One notable observation is the presence of error floors in the 'No Reduction' and DFT-spread curves, where the BER does not decrease significantly despite increasing OSNR. In the 'No Reduction' case, the OFDM signal retains its inherently high PAPR, leading to severe nonlinear distortions in the IQ modulator and optical fiber. These distortions introduce ISI and additional phase noise, which degrade signal integrity even at high OSNR values. Since fiber nonlinearities are power dependent, increasing OSNR does not mitigate their impact, resulting in an increase of the BER and the formation of an error floor. For the DFT-spread technique, although it reduces PAPR to some extent by spreading the energy of subcarriers, inter-band interference among sub-bands in DFT-spread can cause phase noise accumulation, further degrading system performance, as observed in [22]. Moreover, phase noise impacts DFT-spread structures by inducing subcarrier phase errors (SPE). This results in alterations to the phase alignment of subcarriers, which requires the implementation of advanced compensation methodologies to mitigate this impairment [33]. These factors contribute to the observed error floor, where additional OSNR does not significantly improve BER performance.

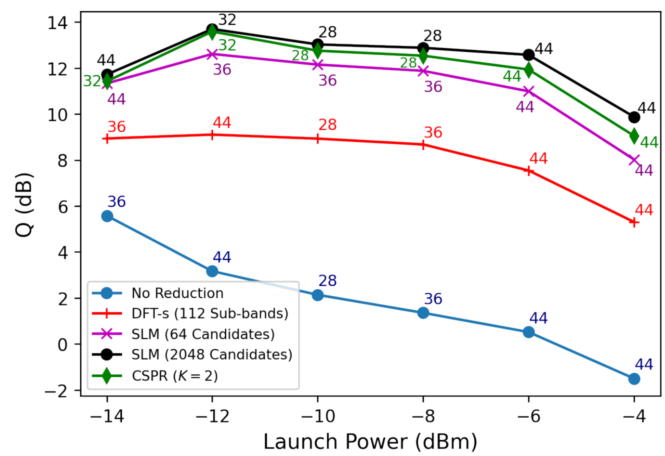

Fig. 12: Launch power vs. Q-factor over 800 km transmission distance

The evaluation of the transmission system for different launch powers over an 800 km transmission distance is shown in Fig. 12. The number next to the point represents the OSNR value at which the Q-factor is measured. It is clear that as the launch power increases, fiber nonlinear effects become more pronounced, resulting in a significant degradation of the Q-factor. The optimal launch power is determined to be approximately −12 dBm, since lower launch powers would lead to amplified spontaneous emission (ASE) generated by EDFA becoming the predominant noise source, thus limiting the performance of the system. A key observation from Fig. 12 is that the proposed CSPR technique consistently achieves a higher Q-factor across all launch power levels compared to DFT-spread and SLM with 64 candidates. With an optimal launch power of −12 dBm, CSPR provides the best system performance, strengthening its effectiveness in mitigating nonlinear impairments. This is mainly attributed to its superior PAPR reduction, which minimizes the impact of nonlinearities of the fiber and reduces the probability that the signal exceeds the nonlinear threshold of the transmission medium. Interestingly, while SLM with 2048 candidates slightly surpasses CSPR in Q-factor performance at its peak, this improvement comes at the cost of significantly higher computational complexity due to the need for additional IFFT computations, search complexity process, and side information transmission.

These results highlight the ability of the proposed CSPR to optimize system performance without requiring additional computational complexity, making it a highly efficient and practical solution for long-haul optical transmission.

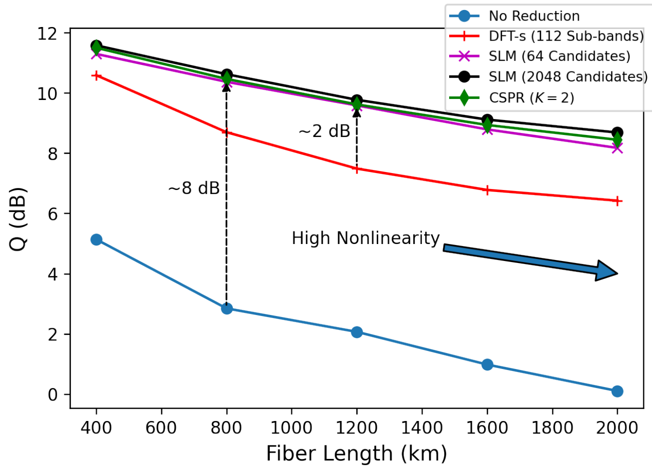

Fig. 13: Fiber length vs. Q-factor at a launch power of -12 dBm

Finally, Fig. 13 illustrates the Q factor of the system as a function of the fiber length at a launch power of -12 dB and an OSNR of 24 dB. As expected, longer transmission distances aggravate fiber nonlinearity, leading to increased Q-factor degradation. The CSPR consistently outperforms the baseline reduction techniques, achieving an 8 dB Q-factor improvement at 800 km. Compared to the DFT-spread, the CSPR maintains a 2 dB Q-factor advantage over 1200 km. Notably, at longer distances where fiber nonlinearity is significant, the CSPR still demonstrates superior performance compared to other techniques. Interestingly, SLM with a large number of candidates (2048) could potentially achieve marginally better Q-factor performance in such extreme scenarios; this comes at the expense of substantial computational overhead and increased system complexity. In contrast, CSPR maintains a favorable trade-off between PAPR reduction and implementation feasibility, making it a highly practical solution for long-haul optical communication systems where both performance and efficiency are critical.

VIII. Conclusion

This work presented CSPR, a groundbreaking PAPR reduction technique examined in CO-OFDM systems. Compared to conventional methods such as clipping, SLM, and DFT-spread, CSPR offers substantial benefits, including a simplified implementation, no reliance on side information, and lower computational requirements. The simulation results underline the effectiveness of CSPR in enhancing system performance. Specifically, CSPR achieves a 3 dB improvement in BER over DFT-spread, on the other hand, CSPR maintains BER error-free at higher OSNR, while the SLM exhibits an error floor. Moreover, CSPR provides an 8 dB Q factor gain at a distance of 800 km with a launch power of -12 dBm, demonstrating its superior handling of fiber nonlinearity and IQ modulator nonlinearity.

Furthermore, CSPR provides computational efficiency compared to SLM and DFT-spread. It requires a computational complexity of $O(N \log C_0)$, which significantly reduces processing overhead without requiring additional complexities, such as side information transmission or search complexity. This makes CSPR an optimal choice for high-performance optical communication systems, where efficient reduction in PAPR is critical to achieving reliable and cost-effective operation.

Future research could investigate the integration of CSPR with emerging optical technologies such as exploring the scalability of CSPR to multi-channel systems such as wavelength division multiplexing (WDM), which would further strengthen its potential for deployment in next-generation optical networks.

Appendix A
RF-Pilot Compensation Concept

In this paper, the RF-Pilot technique addresses laser phase noise by incorporating a pilot tone at the center of the transmitted OFDM symbol. This is achieved by setting the DC subcarrier at zero $N_{DC} = 0$ and introducing a DC offset through the bias voltage of the IQ modulator V_{bias}, at the transmitter, as shown in Fig. 6. The transmitter signal can be expressed as [26], [27], [29], [34]:

$$E_{tx}(t) = A_{data}(t)e^{j(\omega_c t + \phi_{data}(t))} + A_p e^{j(\omega_p t)} \tag{A.1}$$

where A_{data} is the amplitude of the data signal, ω_c is the angular frequency of the optical carrier, and ϕ_{data} is the phase of the data signal. A_p and ω_p represent the amplitude and angular frequency of the pilot tone, respectively. The pilot-to-signal ratio (PSR) significantly impacts the effectiveness of the RF-Pilot technique in mitigating phase noise which is given by [26]:

$$PSR_{dB} = 10 \cdot log_{10}\left(\frac{|A_p|^2}{|A_{data}|^2}\right), \tag{A.2}$$

where $|A_p|^2$ is power of RF-Pilot and $|A_{data}|^2$ is the power of OFDM signal. In alignment with [26] and [35], and simulation measurements, the PSR is determined to be -20 dB in this study. A GB surrounds the pilot at the receiver to improve recovery, but this reduces spectral efficiency due to unused subcarriers. This paper employs a 40-subcarrier GB around the RF-pilot, causing about a 4% bandwidth loss for the 1024-point IFFT. As indicated previously, the received signal $E_{rx}(t)$ with RF-pilot is affected by laser phase noise ϕ_{LPN}.

$$E_{rx}(t) = (A_{data}(t) \cdot e^{j(\omega_c t + \phi_{data}(t))} + A_p \cdot e^{j(\omega_p t)}). \cdot e^{j\phi_{LPN}(t)} \tag{A.3}$$

A bandpass filter centered at the RF-pilot frequency f_p extracts the RF-pilot, yielding the filtered signal E_p.

$$E_p(t) \approx A_p(t) \cdot e^{j(\omega_p t + \phi_{LPN}(t))}, \tag{A.4}$$

then the phase of the extracted pilot is used to estimate the laser phase noise, which is then utilized to mitigate the received data signal. The phase-corrected data signal is

$$E_{rx,corr}(t) = E_{rx}(t) \cdot e^{-j(\arg(E_p(t)) - \omega_p t)}. \tag{A.5}$$

APPENDIX B
IQ MODULATOR MODEL

An IQ modulator uses two Mach-Zehnder modulators (MZMs) to create in-phase (I) and quadrature (Q) components combined with a $\pi/2$ phase shift. The MZMs are driven by baseband signals I_{data} and Q_{data}. The output optical field of an MZM is a cosine function of the driving signal and bias voltage, with its voltage-to-optical field transfer function given by [36]–[38]:

$$E_{MZM}(V(t)) = E_0(t) \cos\left(\frac{\pi V(t)}{2V_\pi}\right), \tag{B.1}$$

where E_0 is the input optical field, $V(t)$ is the control voltage applied to the modulator, and V_π is the voltage required to induce a π phase shift in the modulator. For the in-phase (I) and quadrature (Q) components, the applied voltages are $I_{data}(t)+V_{bI}$ and $Q_{data}(t)+V_{bQ}$, respectively, where V_{bI} and V_{bQ} are the bias voltages. Thus, the total optical field output of the IQ modulator is the combination of these two fields, with a phase shift $\pi/2$ for the quadrature component:

$$E(t) = \frac{E_0(t)}{\sqrt{2}} \left[\cos\left(\frac{\pi(I_{data}(t) + V_{bI})}{2V_\pi}\right) + j \cdot \cos\left(\frac{\pi(Q_{data}(t) + V_{bQ})}{2V_\pi}\right)\right], \tag{B.2}$$

The factor $\frac{E_0}{\sqrt{2}}$ normalizes the input optical field E_0. To incorporate phase noise into the IQ modulator model, phase fluctuations in the optical carrier can be considered. The optical carrier with phase noise can be represented as:

$$E_0(t) = A_i e^{j(w_c t + \phi^{LPN}(t))}, \tag{B.3}$$

where A_i is the input optical field amplitude, w_c the optical carrier's angular frequency, and ϕ^{LPN} the time-varying phase noise modeled as a random walk in Eq. (C.1). V_π is the MZM's half-wave voltage; applying V_π causes a π phase shift. Setting V_π to 2 V positions the MZM near its quadrature point for increased modulation depth. A bias voltage of $V_b = -V_\pi$ centers the MZM's operating point at zero phase shift, allowing efficient modulation of both positive and negative signals [36], [37].

APPENDIX C
LASER PHASE NOISE MODEL

To simulate laser phase noise, random noise samples with a specific variance are added to the oscillator at each time step [30]. The noise accumulated over time emulates the random-walk phase noise process. The discrete-time random-walk model is generated iteratively for N samples as follows:

$$\phi_{i+1}^{LPN} = \phi_i^{LPN} + X, \quad i \in [0, N-1], \tag{C.1}$$

where ϕ_i^{LPN} and ϕ_{i+1}^{LPN} represent the phase noise at the current and next sample, respectively. X is a random variable drawn from a normal distribution with mean μ and standard deviation σ, denoted as:

$$X \sim \mathcal{N}(\mu, \sigma^2). \tag{C.2}$$

The random variable in this paper follows a zero-mean normal distribution with a standard deviation σ, which is related to the laser linewidth Δv and the sampling period T_s as follows:

$$\sigma^2 = 2\pi \cdot \Delta v \cdot T_s. \tag{C.3}$$

APPENDIX D
FIBER LINK MODEL

The Split-Step Fourier Method (SSFM) solves the nonlinear Schrödinger equation (NLSE) that describes the propagation of optical signals in a fiber. The SSFM takes into account the effects of fiber loss, chromatic dispersion, and nonlinear distortions caused by the Kerr effect. The equation that describes this is given by [39], [40]:

$$\frac{\partial A(l,t)}{\partial l} + \frac{\alpha}{2} A(l,t) + j\frac{\beta}{2}\frac{\partial^2 A(l,t)}{\partial t^2} - j\gamma|A(l,t)|^2 A(l,t) = 0, \tag{D.1}$$

where $A(l,t)$ is the envelop of the optical field envelope. l and t represent the propagation distance and the time, respectively. The parameters α, β, and γ model the attenuation, chromatic dispersion, and nonlinear effects, respectively.

The SSFM models fiber propagation by separating linear and nonlinear effects. It implements linear effects like chromatic dispersion (β) of 16 ps/nm/km and attenuation (α) of 0.2 dB/km in the Fourier domain, and nonlinear effects such as the Kerr effect using a coefficient γ of 1.3 $\mathrm{W}^{-1} \cdot \mathrm{km}^{-1}$ in the time domain. The process involves alternating between these domains. Each 80 km fiber span is split into smaller steps of 0.5, enhancing accuracy by minimizing numerical errors. Post propagation, an EDFA with a 4.5 noise figure compensates for 0.2 dB/km attenuation.

APPENDIX E
COHERENT RECEIVER MODEL

Coherent detection mixes the received signal with a local oscillator, generating terms based on the relative phase and frequency difference of the signals. Analyzing these terms allows for the extraction of amplitude and phase information from the received signal. The received optical signal and the local oscillator signal can be represented as [36]–[38]:

$$E_{rx}(t) = A_{rx} e^{j(w_c t + \phi_{rx}(t))}, \tag{E.1}$$

and,

$$E_{LO}(t) = A_{LO} e^{j(w_{LO} t + \phi_{LO}(t))}, \tag{E.2}$$

where A_{rx} and A_{LO} are the amplitudes of the received optical signal and the local oscillator (LO), respectively. The frequencies of the received signal and the LO are w_c and w_{LO}. For simplicity, this paper assumes $w_{LO} = w_c$. The

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phase noise of the received signal, ϕ_{rx}, includes both the laser phase noise of the transmitter and the noise of the transmission medium. The LO laser's phase noise is ϕ_{LO}, modeled as per Appendix C. The mixing process results in

$$I_{rx}(t) = A_{rx}A_{LO}(e^{j(\phi_{rx}(t)-\phi_{LO}(t))} + e^{-j(\phi_{rx}(t)-\phi_{LO}(t))}), \tag{E.3}$$

and

$$Q_{rx}(t) = A_{rx}A_{LO}(e^{j(\phi_{rx}(t)-\phi_{LO}(t))} - e^{-j(\phi_{rx}(t)-\phi_{LO}(t))}), \tag{E.4}$$

where the signal $I_{rx}(t)$ and $Q_{rx}(t)$ are the in-phase component and the quadrature component of the electrical signals received.

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